

Project Documentation

RythmicTunes:Your Melodic Companion

1. Introduction

- **Project Title:** RythmicTunes:Your Melodic Companion
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2. Project Overview:

RythmicTune is a music streaming and discovery platform that provides personalized music recommendations, curated playlists, and social features for music enthusiasts.

• Features:

- Personalized music recommendations based on listening history
- Curated playlists for different moods and activities
- Social features to share music with friends
- Offline listening capability

- Lyrics synchronization

- Artist radio stations

2. Architecture

- **Frontend:** React.js with Tailwind CSS and Framer Motion for animations

- **Backend:** Node.js and Express.js handling music streaming and user management

- **Database:** MongoDB stores user profiles, playlists, listening history, and music metadata

- **Audio Processing:** Custom audio engine for seamless streaming

4. Setup Instructions

- **Prerequisites:**

Node.js (v16 or higher)

- MongoDB (v5 or higher)

- Git

- React.js

- Express.js

- Mongoose

- Visual Studio Code (recommended)

- **Installation Steps:**

Clone the repository

Git clone

<https://github.com/rythmictune/app.git>

Install client dependencies

Cd client

Npm install

Install server dependencies

```
Cd ../server
```

```
Npm install
```

5. Folder Structure

RythmicTune/

```
|-- client/      # React frontend
  |__ components/ # Reusable UI components
  |__ pages/     # Application pages
  |__ hooks/     # Custom React hooks
  |__ utils/     # Helper functions
|-- server/     # Node.js backend
  |__ routes/    # API routes
  |__ models/    # Database models
  |__ controllers/ # Route controllers
  |__ middleware/ # Custom middleware
  |__ utils/     # Server utilities
|-- audio-engine/ # Custom audio processing
|-- docs/        # Project documentation
```

6. Running the Application

- **Frontend:**

```
cd client
```

```
Npm start
```

- **Backend:**

```
Cd server
```

```
Npm run dev
```

- **Access:** Visit <http://localhost:5173/>

7. API Documentation

- **Authentication:**

- POST /api/auth/register

- POST /api/auth/login

- POST /api/auth/logout

- GET /api/auth/refresh
- Users:
 - GET /api/users/profile
 - PUT /api/users/profile
 - GET /api/users/friends
 - POST /api/users/friends
- Music:
 - GET /api/music/songs
 - GET /api/music/songs/:id
 - GET /api/music/artists
 - GET /api/music/artists/:id
 - GET /api/music/albums
 - GET /api/music/albums/:id
- Playlists:
 - GET /api/playlists
 - POST /api/playlists
 - GET /api/playlists/:id
 - PUT /api/playlists/:id
 - DELETE /api/playlists/:id
- Recommendations:
 - GET /api/recommendations
 - GET /api/recommendations/:type

8. Authentication

- JWT-based authentication for secure login
- Refresh token implementation for persistent sessions

- Middleware protects private routes and resources
- OAuth integration for social media login options

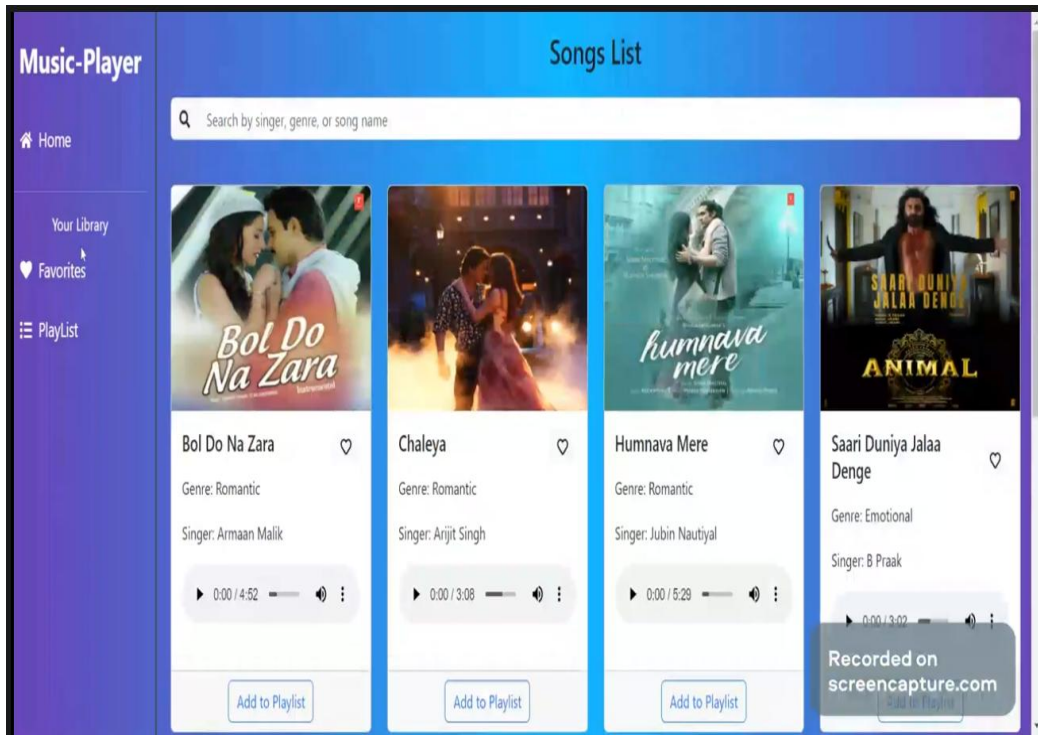
9. User Interface

- Landing Page with featured music and genres
- User Dashboard with recent activity and recommendations
- Music Player with full controls and visualizations
- Search functionality with filters
- User Profile with listening statistics
- Social Feed showing friends' activity

10. Testing

- Unit testing with Jest and React Testing Library
- Integration testing for API endpoints
- End-to-end testing with Cypress
- Manual testing during development milestones
- Tools: Postman, Chrome Dev Tools, Lighthouse

11. Screenshots or Demo



12. Known Issues

Mobile Safari occasionally has audio playback issues

- High memory usage when managing large playlists
- Social notifications sometimes delayed
- Lyrics synchronization can be off by 1-2 seconds for some songs

13. Future Enhancements

- Podcast integration
- Collaborative playlist editing
- Music identification feature (Shazam-like)
- Crossfade between tracks
- Smart alarms with gradual music wake-up
- Integration with smart home devices
- Virtual concerts and live events
- Advanced audio equalizer and sound profiles